PLAIN-LANGUAGE GUIDE · FIRST TIME DOING THIS

Setting Up Your Jetson

Check first — you may not need to flash at all

Tanna TechBiz **Eagle 101** + Jetson Nano · serial 1010425035310

Before anything: what is "flashing"?

Your Jetson is a small computer that probably has no operating system on it yet.

Normally you'd install an OS from a USB stick. The Jetson can't do that. Instead you connect it to **another computer with a cable** and push the operating system across. That is flashing.

Word	What it actually means
Flashing	Installing the operating system onto the Jetson
Recovery mode	A special "listening" mode where the Jetson waits to receive an OS, instead of trying to start its own
BSP	The driver package for *your specific board*. Like laptop drivers — Dell drivers won't work on an HP. Tanna makes the one for the Eagle 101

> **Don't worry about breaking it**
>
> It is very hard to permanently damage a Jetson this way. If flashing fails, you simply do it again. The board almost always goes back into recovery mode. There is only one genuinely delicate moment (Step 10) and it is clearly marked.

> **The most important thing on this page**
>
> **You might not need to flash at all.** Some boards arrive with everything already installed. Part 2 is a 30-minute check that tells you. Do that first — it could save you finding a Linux computer.

Part 1 · What you need

FOR THE CHECK — DO THIS FIRST, TAKES 30 MINUTES

- ☑ The Eagle 101 board with the Jetson module fitted
- ☑ **5 V / 4 A power adapter** with a barrel plug — *not* a phone charger
- ☑ HDMI cable and a monitor or TV
- ☑ A USB keyboard
- ☑ Your Arducam camera

ONLY IF THE CHECK SAYS YOU MUST FLASH

- ☐ A computer running **Ubuntu Linux 18.04 or 20.04**, Intel or AMD — *not* a Mac
- ☐ A **USB to Micro-USB cable** that carries data, not just charging
- ☐ Tanna's BSP files (you have to email them)

☐ About 40 GB free space on that Linux computer

Part 2 · The check

1 Plug things in — in this order

1. HDMI cable to the monitor
2. Keyboard into any USB port
3. Power **last**

Then wait up to **2 minutes**. The very first start-up is slow — don't panic if nothing happens immediately.

2 What do you see on screen?

What happens	What it means
Nothing, and the fan isn't spinning	Check the power adapter, and that the module is pushed fully into its slot
Fan spins, but the screen stays black	No operating system → **you need to flash**
NVIDIA logo, scrolling text, then a login screen	An OS is already installed → **go to Step 3**

3 Only if it started: check the camera

This is the step that really decides it.

Turn the board off. Connect the Arducam to the camera slot. **Turn it back on.** Then open a terminal and type:

```
ls /dev/video0
```

What it prints	What it means
`/dev/video0`	**The correct drivers are installed. You do NOT need to flash.**
`No such file or directory`	Wrong drivers → **you need to flash**

Why the camera decides it: a board can start up perfectly on generic NVIDIA software and still have no camera, because the Eagle 101 wires its camera differently. Only Tanna's BSP knows that wiring.

4 Also check the free space

```
df -h /
```

Write the number down. If less than about 3 GB is free, the board's built-in storage is nearly full, and flashing may be the tidier option even if the camera works.

> **If the camera worked — you're finished with this guide**
>
> No flashing needed. You just need to move the large files onto your 128 GB memory card, which is about five commands. Everything from Part 3 onwards can be ignored.

Part 3 · If you do need to flash

5 Email Tanna — do this before anything else

Everything else waits on their reply, and replies take days. Send this today.

To: `sales@tannatechbiz.com`

```
Subject: Eagle 101 – flashing files needed

Hello,

I have an Eagle 101 Nano Base Full System, serial 1010425035310.

Could you please send me:

  1. The BSP files for JetPack 4.6
  2. The exact board name to use in the flash.sh command
  3. Instructions for putting the board into recovery mode –
     which jumper or pins to connect
  4. Do you offer a flashing service? I would happily send
     the board to you.

Thank you.
```

> **Question 2 is the critical one**
>
> That "board name" tells the installer which drivers to use. If you guess it, you can end up with a board that starts but has no camera — or one that doesn't start at all. **Do not guess it. Wait for their answer.**

And do ask question 4. If they'll flash it for you, everything below disappears.

6 Find a Linux computer

You need **Ubuntu 18.04 or 20.04**, and only for one afternoon.

Options, best first:

1. **A lab computer at your university** — an AI or CS lab almost certainly has one
2. **Install Ubuntu alongside Windows on your i5 PC** (dual boot) — Windows stays untouched
3. Borrow a laptop

Will NOT work	Why
Windows	Cannot build the Linux file system the Jetson needs
Mac (M1 / M2 / M4)	NVIDIA's tools only run on Intel and AMD chips
A virtual machine	The USB connection drops in the middle of flashing

7 Prepare the Linux computer

Open a terminal and paste this in one go. It will ask for your password.

```
sudo apt update
sudo apt install -y qemu-user-static abootimg nfs-kernel-server \
                    libxml2-utils binutils python3 device-tree-compiler \
                    lbzip2 cpio udev dosfstools
```

Let it finish before moving on.

8 Unpack Tanna's files

Follow their instructions — it usually looks like this:

```
mkdir -p ~/nvidia
cd ~/nvidia
tar xf <the-file-Tanna-sent>.tbz2
cd Linux_for_Tegra/rootfs
sudo tar xpf ../../<the-rootfs-file>.tbz2
cd ..
sudo ./apply_binaries.sh
```

That last command takes a few minutes. **It must finish with no errors.** If you see errors, stop and ask for help rather than continuing.

9 Put the Jetson into recovery mode

This is the part that feels strange the first time. Take it slowly.

1. **Unplug the power** from the Jetson
2. Find the **recovery pins**, labelled `FC_REC` and `GND`. Tanna's instructions say exactly where. Connect them with the little plastic jumper, or hold a paperclip across them
3. **Keeping them connected**, plug the power back in
4. Count to five, then remove the jumper or paperclip
5. Connect the **Micro-USB cable** from the Jetson to the Linux computer

Now check it worked. On the Linux computer:

```
lsusb
```

Look through the list for a line containing **NVIDIA** and **APX**.

Result	What to do
You see that line	Continue to Step 10
You don't	Try again. The commonest cause by far is a **charge-only USB cable** — try a different one before anything else

> **Do not run Step 10 until you can see that NVIDIA APX line.**

10 Flash

Replace `BOARDNAME` with the exact name Tanna gave you:

```
cd ~/nvidia/Linux_for_Tegra
sudo ./flash.sh BOARDNAME mmcblk0p1
```

Now wait **10 to 20 minutes**. A lot of text will scroll past — that is completely normal.

> **Do not unplug anything while this runs**
> Not the power, not the USB cable. This is the only genuinely delicate moment in the whole process.

You are looking for:

```
*** The target ... has been flashed successfully. ***
```

11 Start it up

1. Turn the Jetson off
2. Unplug the USB cable
3. Connect HDMI, keyboard and power
4. It should boot into an Ubuntu setup screen — create your username and password

Then check the camera again:

```
ls /dev/video0
```

If that appears, **you're done**.

Part 4 · Straight after flashing

Four things, in this order. Copy each block, press Enter, wait for it to finish.

```
# 1. Turn off the desktop. Frees about 1 GB of memory, which you need later.
sudo systemctl set-default multi-user.target

# 2. Add "extra memory" on the card. Without this, later steps run out of memory
#    and stop with no explanation.
sudo fallocate -l 8G /swapfile
sudo chmod 600 /swapfile
sudo mkswap /swapfile
sudo swapon /swapfile

# 3. Run at full speed
sudo nvpmodel -m 0
sudo jetson_clocks

# 4. Sound support, for the spoken alerts
sudo apt install -y espeak libespeak1 alsa-utils
```

Then restart. After this you can unplug the monitor and keyboard — you'll connect from your Windows PC over the network instead.

If something goes wrong

Problem	Most likely cause
`lsusb` shows no NVIDIA device	**Charge-only USB cable.** Try a different one first — this is by far the commonest cause
Flashing stops partway	Bad cable, or you plugged through a USB hub. Use a port directly on the computer
Flashed fine but won't start	Wrong board name — go back to Tanna
Starts, but no camera	Generic NVIDIA files were used instead of Tanna's BSP
Restarts by itself at random	Power adapter too weak. It needs a real 5 V / 4 A

> **You can always try again**
>
> Just repeat Steps 9 and 10. A failed attempt doesn't damage the board.

Two things to do today

1. **Run Part 2.** Thirty minutes, and it may mean you skip everything else.
2. **Email Tanna.** Their reply takes days, and every later step depends on it.

EchoPath — plain-language Jetson setup guide · prepared 14 August 2026 · unit serial 1010425035310.
Anything specific to your board — the board name, the recovery pin location, the BSP layout — must come from Tanna TechBiz and is marked as such. For the version with full technical detail and exact commands, see *EchoPath_Jetson_BringUp_Runbook.pdf*.